School Effectiveness and School Improvement
An International Journal of Research, Policy and Practice

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/nses20

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SCHOOL EFFECTIVENESS AND SCHOOL IMPROVEMENT
2021, VOL. 32, NO. 4, 509–542
https://doi.org/10.1080/09243453.2021.1906283

The participant effects of private school vouchers around the globe: a meta-analytic and systematic review

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ABSTRACT

School voucher programs are scholarship initiatives – frequently government funded or incentivized – that pay for students to attend private schools of their choice. This study is the first formal meta-analytic consolidation of the evidence from all randomized controlled trials (RCTs) evaluating the student-level math and reading test score effects of school vouchers internationally. Our search process turned up 9,443 potential studies, 21 of which were ultimately included. These 21 studies represent 11 different voucher programs. Utilizing a random effects and robust variance estimation framework, we find moderate evidence of positive achievement impacts of private school vouchers, with substantial effect heterogeneity across programs and outcome years. Suggestive evidence exists that voucher interventions may be cost effective even for null achievement impacts. Future experimental studies of long-term, scaled up voucher interventions could strengthen our understanding of achievement impacts.

ARTICLE HISTORY
Received 15 September 2019
Accepted 10 March 2021

KEYWORDS
Private school vouchers; school choice; systematic review; meta-analysis; opportunity scholarships; achievement effects

Introduction

School vouchers provide resources to families that enable them to attend a private school of choice (Wolf, 2008b). A private school choice initiative is a "voucher program" if the government funds the program directly and the "voucher" is given to a single education provider. Other private school choice initiatives fulfill the same purpose but are funded indirectly, through tax credits to businesses or individuals who contribute to nonprofit scholarship organizations. Private school choice programs are common around the world (e.g., Glenn et al., 2012; Wolf & Macedo, 2004). Such programs are either universal or targeted. Universal programs offer funding to all school-age children in a jurisdiction, whereas targeted programs limit private school choice to students with disabilities or who come from low-income families.

Universal private school choice programs operate in Belgium, Denmark, the Netherlands, Sweden, and other European and Commonwealth countries mainly based on a

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Supplemental data for this article can be accessed at https://doi.org/10.1080/09243453.2021.1906283
This article has been republished with minor changes. These changes do not impact the academic content of the article.

constitutional right for parents to educate their children in a particular religious, philosophical, or pedagogical tradition (Glenn, 1989). A universal school voucher program has operated in Chile since the 1980s (Mizala & Romaguera, 2000). Of the 54 private school choice programs in 28 US states plus the District of Columbia, nine are nearly universal in design, only limited due to funding amounts (EdChoice, 2017). Those programs are in Arizona (two), Georgia, Montana, and Nevada; the rural areas of Maine, New Hampshire, and Vermont; and Cleveland, Ohio.

Private school choice programs targeted to low-income students operate in Colombia, India, Pakistan, and several African countries. They provide around \$200 per year per student to attend low-cost private schools (Dixon, 2013; Tooley, 2009). The US hosts 45 targeted choice programs: 23 are means tested, 17 are limited to students with disabilities, one is restricted to students attending failing public schools, and four are doubly targeted to low-income students in low-performing schools (EdChoice, 2017). Scholarships range from \$1,000 to \$13,000. The lower cost programs typically require families to pay extra. Vouchers for students with disabilities tend to be larger, cover the full cost of education, and sometimes are valued based on the severity of the disability.

Voucher supporters claim that participating students will learn more by accessing higher quality schools or by being better matched to a school that meets their needs (DeAngelis & Erickson, 2018). School voucher benefits are fiercely contested (Doolittle & Connors, 2001; McEwan, 2000). Ravitch (2014) calls school vouchers a "hoax" that fails kids. Some of the common arguments against vouchers are that they lead to sorting and stratification of students by income and ability, and that they "cream skim" the best students from the public schools (Elacqua, 2012; Epple et al., 2017). Murnane (2005), in contrast, argues: "Providing families who lack resources with educational choices makes sense. The consequences of attempting to do this through a large-scale voucher … system are unknown. Carefully designed experiments could provide critical knowledge" (p. 181).

Experimental design is critical to evaluating school voucher programs due to concerns about selection bias. We expect more motivated and able families to self-sort into private schools and/or voucher programs (Fleming et al., 2015), though that is not always the case (K. P. Anderson & Wolf, 2017). Fortunately, much of the school voucher research in the US, as well as some of it abroad, has involved experiments that control for self-selection in expectation. Well-executed random assignment ensures that access to private schooling is due to chance, not selection, ensuring that experimental results are causal. However, applicants to voucher programs may be motivated differently than the broader set of families who could access choice at scale. Similarly, private schools that participate in voucher programs may be different from private schools that do not. Both of these factors limit the external validity of experimental voucher findings.

This study is the first global meta-analysis of all randomized controlled trials (RCTs)[1] of voucher math and reading impacts on students publicly reported through 2018. Our search turned up 9,443 potential studies, 21 of which were included representing 11 different programs, eight in the US and three elsewhere. We focus on math and reading achievement because schools are charged with enhancing student learning, so test-score outcomes are a reasonable measure of program impacts, and math and reading effects of private school choice programs are common in the literature.

Very few voucher RCTs have evaluated non-math and non-reading test score or non-achievement outcomes. Results suggest school choice has negative or null effects on science and social studies scores (Abdulkadiroglu et al., 2015; Mills & Wolf, 2016, 2017; Muralidharan & Sundararaman, 2015) but positive effects on educational attainment (Angrist et al., 2002; Wolf et al., 2013). Voucher lottery winners are less likely to marry or cohabit as teenagers (Angrist et al., 2002), and more altruistic towards charitable organizations (Bettinger & Slonim, 2006). Although these effects on outcomes other than math and reading achievement are interesting, there are too few of them to consolidate into a formal meta-analysis without introducing a substantial amount of statistical noise.

Prior systematic reviews of voucher effectiveness

Private school choice is increasingly common throughout the US[2] and the world. The US and its territories hosted 56 different private school choice programs serving over 500,000 students in 2019–2020 (Wolf, 2020). Nearly half of the 53 countries the Organisation for Economic Co-operation and Development (OECD, 2013) surveyed had policies or programs providing public money in support of students enrolled in private schools. Multiple scholars have reviewed research on the effectiveness of voucher programs over the past decade. We review the systematic reviews of voucher effectiveness to evaluate if they reveal a clear, consistent, and indisputable judgment on the math and reading achievement effects of vouchers.

From 2008 through 2018, 15 reviews of the achievement effects of private school choice have been published as reports, working papers, or journal articles. This plethora of reviews underscores the topic's salience. A meta-analysis is a statistical method that combines evidence from studies on a topic in order to develop an overall conclusion on an intervention's effectiveness. Without methodological coherence across the studies, one cannot tell if findings vary because the intervention had heterogeneous effects or because some research designs were biased. The ideal meta-analysis includes studies with similar methodologies, is comprehensive, and provides an estimation of the average effect of an intervention on an important outcome (Hedges & Olkin, 1985; Hunter & Schmidt, 1990; Rossi et al., 2004, pp. 324–328).

Private schools account for a fifth of primary school enrollment in developing countries (Baum et al., 2014). The private school sector has grown in the last 2 decades in the developing world, despite near-universal access to free public primary schools and increased government spending on public education (Glewwe & Muralidharan, 2016). An ideal meta-analysis on the effectiveness of private school vouchers should incorporate studies from around the globe, which is our fourth desirable criterion. None of the 15 existing reviews satisfies even three of the four criteria for an ideal meta-analysis. Seven satisfy none of them (Table 1).

Only seven reviews are methodologically coherent, focusing on experiments (Egalite & Wolf, 2016; Forster, 2011, 2013, 2016; Fryer, 2017; Morgan et al., 2015; Wolf, 2008a). Only two reviews (Egalite & Wolf, 2016; Forster, 2016) include all studies available at the time that fit the authors' inclusion criteria (see Online Appendix 1). Only two reviews (M. R. Anderson et al., 2013; Fryer, 2017) are formal meta-analyses that include overall effect estimates and confidence levels. The other 13 studies are literature reviews. The M. R. Anderson et al. (2013) meta-analysis includes 17 US school voucher studies of widely varying methodologies. The Fryer (2017) meta-analysis is limited to experiments but mixes RCTs of public school choice with those of private school choice. Thirteen

Table 1. Extent to which previous voucher reviews satisfy the conditions for an ideal meta-analysis.

Review	Methodo-logically Coherent	Compre-hensive	Specifies Average Effect	Global	Conclusions
Miron et al. (2008)					Null to positive, encouraging
Lubienski & Weitzel (2008)					Null to positive, discouraging
Wolf (2008a)	X				Null to positive, encouraging
Rouse & Barrow (2009)					Null to positive, encouraging
Coulson (2009)				X	Null to positive, encouraging
Usher & Kober (2011)					No effect
Forster (2011)	X				Consistently positive
Forster (2013)	X				Consistently positive
M. R. Anderson et al. (2013)			X		Null to positive, discouraging
Epple et al. (2017)				X	Null to positive, encouraging
Morgan et al. (2015)	X				Consistently positive
Forster (2016)	X	X			Consistently positive
Lubienski & Brewer (2016)					Null to positive, discouraging
Egalite & Wolf (2016)	X	X			Null to positive, encouraging
Fryer (2017)	X		X		Null to positive, encouraging

Note: An "X" indicates that the condition is satisfied.

reviews have geographic limits. Eleven are limited to the US, Morgan et al. (2015) to developing countries, and Fryer (2017) to developed countries. Epple et al. (2017) and Coulson (2009) are the only reviews including both US and non-US studies.

Previous reviews vary greatly in their conclusions regarding voucher effectiveness. School vouchers have no effect on student achievement (Usher & Kober, 2011), consistently improve it (Forster, 2011, 2013, 2016; Morgan et al., 2015), or produce a mix of null to positive effects that are either encouraging (Coulson, 2009; Egalite & Wolf, 2016; Epple et al., 2017; Fryer, 2017; Miron et al., 2008; Rouse & Barrow, 2009; Wolf, 2008a) or discouraging (M. R. Anderson et al., 2013; Lubienski & Brewer, 2016; Lubienski & Weitzel, 2008). Most of the individual studies in the reviews are only modestly powered to detect voucher effects, having analytic samples of less than 1,000 students in the final evaluation year. The many findings of "no significant effects" could be due to limited power, noisy data, or the absence of a true effect. None of the reviews includes recent studies from 2017 and 2018 that have generated extensive policy and media interest. Given the lack of any coherent, complete, statistical meta-analysis of the effect of private school vouchers on student achievement globally, this study offers a clear contribution to the literature on private school choice.

Method

Our statistical meta-analysis employs a random effects (Borenstein et al., 2007) and robust variance estimation (RVE) framework with small sample corrections (Tipton, 2015) to support the accuracy and interpretation of our results.

Identification and selection of studies

We identified studies using search, inclusion, and exclusion criteria (see Online Appendix 1). We focused on RCT (i.e., experimental) studies because they are the "gold standard" of program evaluation in assessing causality (Mosteller & Boruch, 2002; Pirog et al., 2009; Rossi et al., 2004). Voucher studies that use quasi-experimental methods such as regression discontinuity design, propensity score matching, or control variables to reduce selection bias reveal a mix of results ranging from positive in some subject areas and years (Witte et al., 2014) to consistently null (Witte, 2000) to negative in some subject areas (Figlio & Karbownik, 2016). While well-designed quasi-experimental studies can approximate the results of RCTs under certain conditions (Bifulco, 2012; Fortson et al., 2012), sometimes they fail to reveal the true causal effects of programs (K. P. Anderson & Wolf, 2017; Betts et al., 2010; Cowen, 2008). Often we cannot even tell how much bias is present when relying on quasi-experimental approaches (Pirog et al., 2009). Quasi-experimental approaches may have greater external validity than experimental ones, however, because they are less reliant on distinctive randomized samples of students in a single location. Our study compensates at least partially for the limited external validity of individual voucher experiments by systematically compiling the experimental results across all of them.

Compliance in voucher RCTs refers to program participants remaining in their assigned treatment or control groups. Treatment non-compliance involves students offered vouchers who decline to use them. Control crossover refers to students assigned to the control group who access the treatment group experience of private schooling. RCTs are more reliable under conditions of strong compliance, as treatment decliners or control crossovers reintroduce selection bias if analysts merely compare the outcomes from voucher users to the results of those who remained in public schools. Treatment on the treated (TOT) estimation methods recover unbiased estimates of the effect of actually experiencing the intended school voucher treatment when some treatment and control students are non-compliers. The three TOT strategies used in voucher RCTs are calculation of the complier average causal effect (Cowen, 2008), instrumental variables analysis (Heckman, 1996), and the Bloom adjustment (Bloom, 1984). All three approaches compare the average outcomes for program participants to the outcomes those same students would have experienced if they did not participate in the program, statistically adjusting in one of three ways for how non-compliance has diluted any difference in average outcomes across the experimental groups. Voucher RCTs that lacked the information required to evaluate the TOT method (Greene et al., 1999; Rouse, 1998) were excluded from the meta-analytic calculations. We calculated the TOT manually, using the Bloom adjustment, from voucher RCTs lacking TOT estimates but including sufficient information for us to do so.

We also focus on RCTs because the conclusions of M. R. Anderson et al. (2013) about the efficacy of vouchers varied based on the research methodology of the studies. If one has to believe either RCTs or non-RCTs regarding the effects of a given intervention, then one should believe the results from RCTs because they have stronger internal validity. Finally, we were confident we would harvest enough voucher effect estimates from RCTs globally to produce reliable meta-analytic estimates.

Inclusion and exclusion criteria

To be included, a study had to be an English-language experimental examination of the impact of private school choice on student-level test scores in math, reading, or both. Only studies of the participant effects of voucher programs were included. Studies were not limited by their publication status or date. Twenty-one RCTs met these qualifications. Most of the included studies have been published in peer-reviewed journals, have been released as reports by the Institute of Education Sciences (Dynarski et al., 2017, 2018), or have been printed as a book chapter (Wolf et al., 2015). Two working papers that were subsequently published in journals (Abdulkadiroglu et al., 2015; Mills & Wolf, 2017) inform the test score impacts of the Louisiana Scholarship Program. When necessary, details were added through supporting documents or contact with the authors. Most of the articles excluded were theoretical discussions or opinion pieces without quantitative evidence, were focused on other issues such as competitive or fiscal effects of vouchers, or were merely quasi-experimental. A surprising number of education evaluations described as "experimental" via keywords or abstracts, upon a closer reading, did not create their comparison groups via random assignment, and therefore were excluded as merely quasi-experimental.

Search strategy

We identified publications from systematic computer and networked searches of EBSCO, JSTOR, ProQuest, Google Scholar, and a daily Google alert list (see Online Appendix 3). We also searched the websites of organizations focused on school vouchers: the University of Chile, Uppsala University in Sweden, the Poverty Action Lab at Massachusetts Institute of Technology (MIT), the National Bureau of Economic Research (NBER), the Department for International Development (DFID) in the UK, and the Center for Civil Society in India. Lastly, we consulted subject-matter experts in the field to find additional relevant studies that the systematic search missed. The initial search process began in mid-2015 and yielded 16 qualified RCTs. Network searches found the remaining five studies.

Search results

The search produced 9,443 articles, of which 21 met our inclusion criteria. Two team members reviewed each of the 9,443 sources based on its title and abstract in order to decide if it merited a full review. After our title/abstract review, 276 sources remained. Two team members have reviewed the full version of each of these 276 studies to determine if each one met our inclusion criteria. In the few cases in which the members initially disagreed on inclusion, they met to reach a consensus. We excluded 260 papers at the full article phase (see Figure 1 and Online Appendix 4).

Our search uncovered several voucher RCTs that prior systematic reviews did not examine. The global scope of our search added key data to our study. Two studies of a large voucher program in Bogota, Colombia (Angrist et al., 2002, 2006), and two studies of different programs in separate regions of India (Muralidharan & Sundararaman, 2015; Wolf et al., 2015) have been added. Our combined computer and network search also identified an RCT of a small voucher program in Toledo (Bettinger &

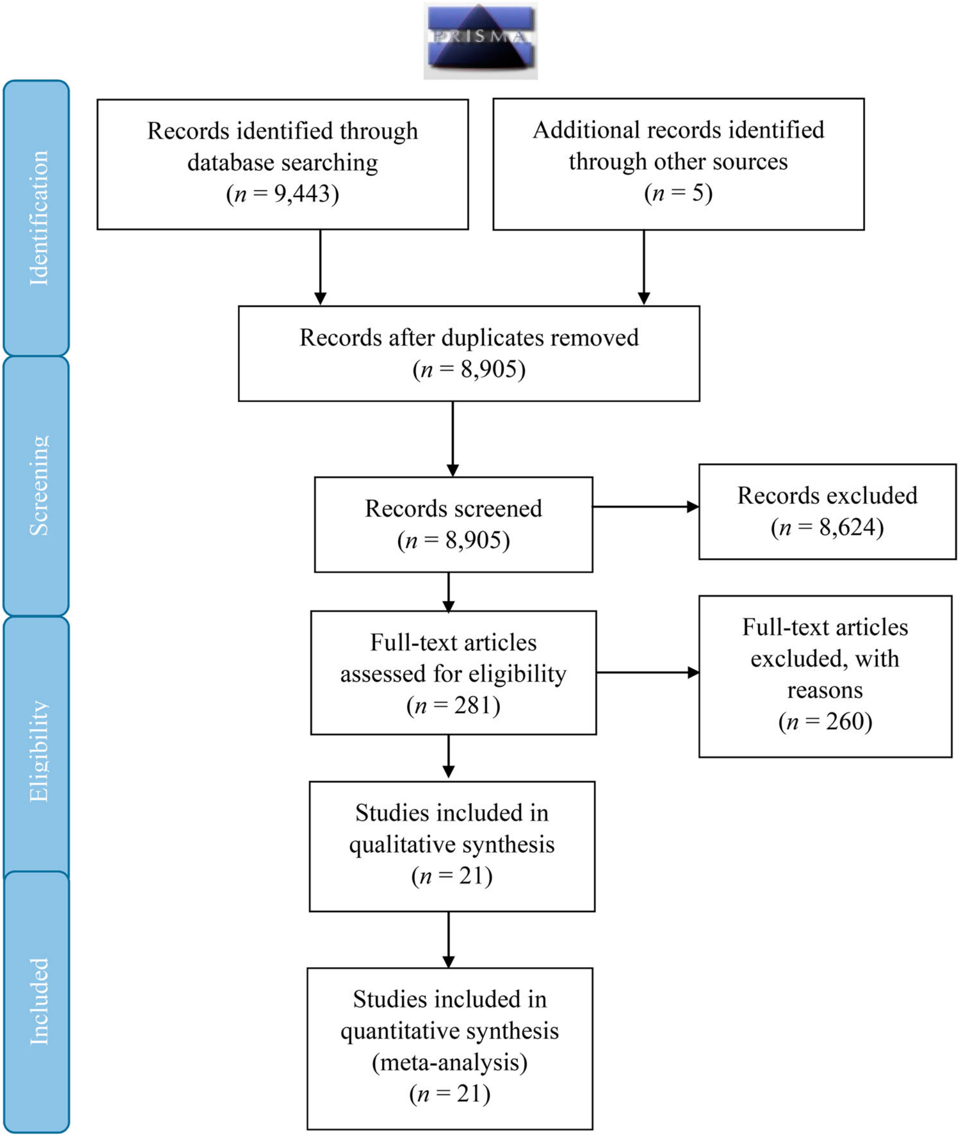

Figure 1. PRISMA flow diagram.

Note: The PRISMA flow diagram shows the stages of identification, selection, eligibility, and inclusion of studies in the meta-analysis. It is based upon the format in Moher et al. (2009).

Slonim, 2006) that all previous reviews missed. Our network search independently identified two evaluations of the Louisiana voucher program (Abdulkadiroglu et al., 2015; Mills & Wolf, 2017), one evaluation of a philanthropic voucher program in Delhi, India (Wolf et al., 2015), and two evaluations of the federal voucher program in Washington, DC (Dynarski et al., 2017, 2018). These studies were not publicly available during the initial phase of our computerized search. This meta-analysis represents a new look at a more comprehensive body of rigorous research on private school vouchers than ever before.

Coding of studies

The 21 included studies were coded using a predesigned format in Microsoft Excel for details on the program evaluated, its location, its authors, its publication year, its funding type (public/private), its years of evaluation, its duration of study, its grades analyzed, its math or reading[3] outcome, its sizes of treatment and control groups, its overall sample size, and its rates of program and sample attrition. Bettinger and Slonim (2006) had only math outcomes, whereas the other 20 studies included both math and reading. Some studies had multiple evaluation years. Louisiana used a placement lottery, so it only contributes to TOT estimates. Each evaluation year, estimator (TOT or intent to treat [ITT]), and subject combination has been treated as a separate observation. A study that reported 3 years of TOT and ITT results in both reading and math contributed 12 observations to the database (3 x 2 x 2). When the authors provided results from multiple estimation models or from robustness checks, we only extracted the estimates from the most robust model that the authors signaled.

Six different research teams published 24 reports or articles analyzing the experimental data from the New York City Children's Scholarship Fund evaluation, 1998–2002. Including all of those reports would bias the meta-analysis by repeatedly counting the same finding. Thus, we treated as a single study any group of publications of the same results, using the same methods, by essentially the same research team. Any variation on that standard, such as publication of different results, using the same methods, by a different research team, represented a different study even though it drew upon the same data. That determination reduced the number of New York City studies to five (Barnard et al., 2003; Bitler et al., 2015; Jin et al., 2010; Krueger & Zhu, 2004; Peterson et al., 2003). We drew data from the final publication unless an earlier one contained more complete statistics, and supplemented the data with descriptive information from other studies of the same program as needed. A study in our meta-analysis is the final, most complete presentation of a specific set of findings from a specific research team using a particular analytic method.

Mean test scores (treatment and control), differences in means (treatment – control), standard deviation (treatment and control), treatment effect size (in standard deviations), standard error of Cohen's *d*, treatment effect and standard error in National Percentile Rank (NPR), *t* statistics, *p* values, and 95% confidence intervals have been entered into the Excel sheet. We used the information to calculate study effect sizes using the formula listed in Appendix 1.

Table 2 summarizes the studies, presenting attrition rates in terms of both sample attrition (the percent of study participants not observed in the final year) and program attrition (the percent of students offered a voucher not using it in the final year). The program attrition rate is the additive inverse of the voucher use rate in the final year, also reported when available. Table 2 indicates how the TOT effects were estimated. Each study randomized at the child or family level.[4]

Table 2 includes the control crossover rate. In the US programs, some students who lost the voucher lotteries found other ways to access private schools. In the RCT in Dayton, Ohio, 18% of the control-group students enrolled in a private school without the aid of a voucher (Howell & Peterson, 2006, p. 44). In the first evaluation of the District of Columbia Opportunity Scholarship Program (DC OSP), 12% of the students who lost the lottery

Table 2. Description of 21 RCT studies included in meta-analysis.

Program Evaluated	Publication	Years of Treatment	Study Duration	Grades	Sample Size (First Outcome Year)	Program Attrition (Final Year)	Sample Attrition (Final Year)	Control Crossover Rate	Voucher Use Rate	TOT method
Andhra Pradesh (AP) School Choice Experiment	Muralidharan & Sundararaman (2015)	4	2008–2012 (4 years)	1 to 5	4,620	49%	20.7% English; 68.1% Hindi; 17.5% Telugu; 17.5% Math	Not provided	61% (initial) or 51% (final year)	Original research team used Bloom (1984) adjustment
Charlotte Children's Scholarship Fund	Greene (2000)	1	1999–2000 (1 year)	2 to 8	357	51.60%	60%	Not provided	48.40%	N/A
	Cowen (2008)	1	1999–2000 (1 year)	2 to 8	347	25.50%	70%	Not provided	74.50%	Complier average causal effect (CACE), the mean treatment outcome across compliers
Children's Scholarship Fund (Toledo, OH)	Bettinger & Slonim (2006)	3	1998–2001 (4 years)	K to 8	186	43%	92%	21% (at time of survey) but 39% of control units attended private school at some point post-lottery	57% (at time of survey) but 65% of winners attended private school at some point post-lottery	Meta-analytic team estimated TOT using Bloom (1984) adjustment
District of Columbia Opportunity Scholarship Program (OSP)	Wolf et al. (2013)	4	2004–2009 (6 years)	K to 12	1,649	28.7%	37.8% Treatment, 48.5% Control	23.10%	74.3% (1st year), 71.3% (final year)	Original research team used Bloom (1984) adj.
	Dynarski et al. (2017)	1	2012–2015 (3 cohorts, 1 outcome yr. each)	K to 12	1,077	30%	24% Reading; 25% Math	10% (1st year)	70% (initial)	Original research team used Bloom (1984) adj.
	Dynarski et al. (2018)	2	2012–2016 (3 cohorts, each with	K to 12	1,001	41%	29% overall (23% in treatment	10%	59%	Original research team used Bloom (1984) adj.

(*Continued*)

Table 2. Continued.

Program Evaluated	Publication	Years of Treatment	Study Duration	Grades	Sample Size (First Outcome Year)	Program Attrition (Final Year)	Sample Attrition (Final Year)	Control Crossover Rate	Voucher Use Rate	TOT method
			two outcome years)				group, 36% in control group)			
Ensure Access to Better Learning Experiences (ENABLE)	Wolf et al. (2015)	2	2011–2013 (2 years)	K to 2	1,306	11%	N/A	25% in total (or 30% of controls with outcome data)	89% (among students with 2nd-year outcome data)	Instrumental Variables analysis
Louisiana Scholarship Program (LSP)	Abdulkadiroglu et al. (2015)	1	2012–2013 (1 year)	3 to 8	N/A	27%	N/A	5%	73%	Instrumental Variables analysis
	Mills & Wolf (2017)	3	2012–2015 (3 years)	3 to 8	1,184	46%	10%	6% (Year 1), 15% (Year 2), 14% (Year 3)	80% (Yr 1), 64% (Yr 2), 54% (Yr 3)	Instrumental Variables analysis
Milwaukee Parental Choice Program (MPCP)	Rouse (1998)	4	1990–1994 (5 years)	K to 8	1,343	64.7%	N/A	Not provided	86.5% (initial fall), 35.3% (spring of final year)	N/A (IV analysis included in paper not comparable to other studies)
	Greene et al. (1999)	4	1990–1994 (5 years)	K to 8	816	N/A	60% Treatment, 52% Control	Not provided	Not provided	N/A (TOT in paper was non-experimental)
Parents Advancing Choice in Education (Dayton, OH)	Peterson et al. (2003)	2	1998–2000 (2 years)	K to 12	404	N/A	51%	18% (1st year) or 10% (both 1st and 2nd years)	78% (1st year), 60% (2nd year)	Instrumental Variables analysis
Programa de Ampliacion de Cobertura de la Educacion Secundaria (PACES)	Angrist et al. (2002)	3	1995–1999 (4 years)	6 to 9	283	43%	75.30%	Majority of controls went to private school at least 1 year*	By survey year, about 57% still using voucher**	Meta-analytic team Bloom (1984) adjusted
	Angrist et al. (2006)	7	1994–2001 (8 years)	6 to 11	3,541	50% (after 3 years)	12.40%	Majority of controls went to private school at least 1 year*	50% after 3 years	Meta-analytic team Bloom (1984) adjusted
School Choice Scholarships	Barnard et al. (2003)	1	1997–2000 (4 years)	1 to 4	525	20% to 27% (overall,	22.30%	6% to 10%***	73% to 80%***	CACE

Foundation Program (NYC)						not final year)				
	Peterson et al. (2003)	3	1997–2000 (4 years)	1 to 4	1,434	30%	33%	5% (1st year) or 3% (both 1st and 2nd years)	82% (1st year), 79% (both years), 70% (final year)	Instrumental Variables analysis
	Krueger & Zhu (2004)	3	1997–2000 (4 years)	K to 4	2,080	41.3% (assumed same as Bitler et al., 2015)	36.20%	11% of control students attended private school in at least 1 year	77% used voucher at least 1 year	Instrumental Variables analysis
	Jin et al. (2010)	1	1997–2000 (4 years)	1 to 4	525	20% (overall, not final year)	22.30%	10%	About 80% overall	CACE
	Bitler et al. (2015)	3	1997–2000 (4 years)	K to 4	2,080	41.3%	34.6% Reading; 35.0% Math	5% (1st year), 6% (2nd year), 8% (3rd year)	74% (1st year), 65% (2nd year), 55% (3rd year)	Meta-analytic team Bloom (1984) adjusted
Washington Scholarship Fund (WSF)	Peterson et al. (2003)	3	1998–2001 (3 years)	K to 8	930	71%	40%	11% (1st year), 8% (both 1st and 2nd years)	68% (1st year), 47% (2 years), 29% (final year)	Instrumental Variables analysis

Notes: The sample size and attrition rates are based on the estimates from Intent to Treat (ITT) Reading except for Bettinger and Slonim (2006), which had only math impacts. The actual sample sizes for calculating the ITT and Treatment on the Treated (TOT) Reading and Math impacts may differ slightly. RCT = randomized controlled trial; K = kindergarten; IV = instrumental variable.
*Voucher eligibility was conditional on admission to a participating school.
**The 57% not using the voucher is based on 15% not in school, 16% in public school, and 12% that lost the voucher for other reasons.
***Control crossover and take-up rates from Barnard et al. (2003) are reported for single-child families, depending on time of application and background strata.

subsequently enrolled in a private school (Wolf et al., 2013, p. 257). In the second DC OSP study, after 1 year, 10% of control-group students attended private schools (Dynarski et al., 2017). The New York City program achieved the clearest treatment–control contrast in type of school attended, as only 4% of the lottery losers attended a private school (Howell & Peterson, 2006, p. 44). In the RCTs in this meta-analysis, students remained in the control group and their outcomes counted towards the control-group average for the ITT impact estimates, even if they attended a private school. The rates at which control-group students crossed over to private schooling factored into the TOT effect calculations (see Online Appendix 2). As such, the ITT estimates are typically lower than the TOT estimates within a study, since the control crossovers reduce the purity of the treatment–control contrast, but not all studies provided ITT impacts (e.g., the Louisiana Scholarship Program was a placement lottery, and it has only TOT effects).

Programs included in meta-analysis

The 21 RCTs identified by our search represent 11 school voucher programs (Table 3). Five programs – in Andhra Pradesh and Delhi, India; Toledo and Dayton, Ohio; and, the DC

Table 3. Description of 11 voucher programs included in meta-analysis.

Program Evaluated	Location	Funding Source	Funding Amount	Grades	Studies Cited
Andhra Pradesh (AP) School Choice Experiment	Andhra Pradesh, India	Private	Full	1 to 5	Muralidharan & Sundararaman (2015)
Charlotte Children's Scholarship Fund	Charlotte, NC (USA)	Private	Partial	2 to 8	Cowen (2008); Greene (2000)
Children's Scholarship Fund	Toledo, OH (USA)	Private	Partial	K to 8	Bettinger & Slonim (2003, 2006)
District of Columbia Opportunity Scholarship Program (OSP)	Washington, DC (USA)	Public	Full	K to 12	Dynarski et al. (2017, 2018); Wolf et al. (2010, 2013)
Ensure Access to Better Learning Experiences (ENABLE)	Delhi, India	Private	Full	K to 2	Wolf et al. (2015)
Louisiana Scholarship Program (LSP)	Louisiana (USA)	Public	Full	3 to 8	Abdulkadiroglu et al. (2015); Mills & Wolf (2016, 2017)
Milwaukee Parental Choice Program (MPCP)	Milwaukee, WI (USA)	Public	Full	K to 8	Greene et al. (1999); Rouse (1998)
Parents Advancing Choice in Education	Dayton, OH (USA)	Private	Partial	K to 12	Peterson et al. (2003)
Programa de Ampliacion de Cobertura de la Educacion Secundaria (PACES)	Bogota, Colombia	Public (partly funded by World Bank)	Partial	6 to 9 (2002); 6 to 11 (2006)	Angrist et al. (2002, 2006)
School Choice Scholarships Foundation Program	New York, NY (USA)	Private	Partial	1 to 4	Barnard et al. (2003); Bitler et al. (2015); Jin et al. (2010); Krueger & Zhu (2004); Peterson et al. (2003)
Washington Scholarship Fund	Washington, DC (USA)	Private	Partial	K to 8	Peterson et al. (2003)

Note: Studies do not necessarily contain all years of a program. See Table 2 for more details at the study level.

Washington Scholarship Fund[5] – were each subject to a single experimental evaluation. Four programs – in Charlotte, NC; Louisiana; Milwaukee, WI; and, Bogota, Colombia – were each the focus of both an original RCT and one replication study. The New York City program has been the subject of five different analyses. The DC OSP has been the subject of two different evaluations.

Each program is categorized as privately or publicly funded, with full or partial vouchers. Participating private schools must accept full vouchers as the total cost of educating the student while partial vouchers require an extra payment from the student's family. In general, the full vouchers studied here were publicly funded and partial vouchers were privately funded, with implications for the types of families who might access and choose to use them. Funding for the vouchers in India and Colombia, whether full or partial, was extremely low in nominal US Dollars, ranging from $117 in India to $190 in Colombia (Angrist et al., 2002; Wolf et al., 2015), reflective of lower educational costs in these areas. Average costs of public schooling are $350 in Colombia (Angrist et al., 2002, pp. 1537–1538) and $1,963 in Delhi, India (Government of Delhi, 2017). For Andhra Pradesh, Muralidharan and Sundararaman (2015, pp. 1031, 1058) report the voucher amount was 40% of the average public school cost per child. The full programs in the US provided vouchers ranging from $5,000 in Louisiana to over $13,000 for high school students in DC (Dynarski et al., 2017; Mills & Wolf, 2016). In both DC and Louisiana, voucher-participating schools were required to accept these funds as the full cost of educating the student, regardless of whether or not their tuition rates typically exceeded these amounts. All private programs in the US were partially funded, and only provided families with about $2,000 in tuition support (Peterson et al., 2003), which then family contributions and school-based financial aid would have to supplement, in order to cover the full tuition. The maximum voucher amount for all the programs in this meta-analysis generally represented less than half of the amount spent per pupil on students in area public schools, as detailed later in the section on cost effectiveness.

All programs were targeted to low-income students through income limits or program location, but usually both. The voucher initiatives in India and Colombia served students living in abject poverty (Angrist et al., 2002; Muralidharan & Sundararaman, 2015; Wolf et al., 2015). The US programs were limited to students whose family incomes were near or below the cut-off for the federal lunch program, almost entirely in cities. Almost all US voucher participants were either African American or Hispanic. This meta-analysis studies the test-score effects of low-cost private school vouchers on low-income, primarily urban, children of color.

Sixty-nine percent of private schools in the DC OSP had full tuition rates of $7,500 or less (Wolf et al., 2007, p. 17), and the average tuition for LSP schools was $5,389 (M. H. Lee et al., 2020). Most of these private schools had experience serving disadvantaged students (Sude et al., 2018). Religious schools generally, and Catholic schools particularly, were the main participants in US voucher programs. In the first evaluation of the DC OSP, 80% of the voucher users attended a religious school and 53% of those students specifically attended a Catholic school (Wolf et al., 2013, p. 257). In US Catholic schools, tuition may only cover about 80% of the budget, with the remainder coming from donations (Gjelten, 2020).

Background and moderator variables

We coded six moderator variables that could indicate effect heterogeneity: years treated, subject tested (reading or math), policy parameter (TOT or ITT), funding source (Private or Public), location (US or International), and other special program characteristics. Years treated varies both within and between programs, while the other moderators vary only between them.

Some studies focus on the TOT effect of using a voucher to attend a private school. Expanding access to students who would not normally have the opportunity to attend a private school and creating competitive pressure for public schools to improve and/or become more efficient are some of the stated purposes of voucher programs. Thus, the participant and competitive effects of actual student transfers into private schools with the support of vouchers are often viewed as the most policy-relevant findings (Bifulco, 2012; Cowen, 2008; Egalite & Mills, 2021; Howell et al., 2002). Other analysts favor the ITT effect of the mere offer of a voucher, as it is the pure experimental impact from the evaluation.

The literature on RCTs in developing countries highlights that often programs that are effective when non-government organizations operate them produce null impacts when governments run them (Glennerster, 2017). Some researchers argue that regulations in government-run voucher programs in the US restrict the participation of quality private schools (Sude et al., 2018). Thus, privately funded programs may yield a higher impact than publicly funded ones, which we define as those with *any* government funding.

The counterfactual condition for control-group students varied across programs. In India and Colombia, almost all the students who lost the voucher lotteries attended local government-run schools. In India, public schools have many more resources than low-cost private schools, but daily teacher absenteeism rates of around 30% plague them, generally without the use of substitute teachers (PROBE Team, 1999). A more recent analysis has revealed the nationwide teacher absence rate in public schools in India to be 23.6%; it costs the nation $1.5 billion annually (Muralidharan et al., 2017). Nationally representative surveys reveal teacher absenteeism in India to be lower in the private schools (Davies, 2019; Kremer et al., 2005), a condition confirmed in one of the school voucher studies in our analysis (Muralidharan & Sundararaman, 2015). The counterfactual condition for students participating in voucher programs in the US is that public schools are free and usually residentially assigned. The public schools in the US face fewer problems with teacher absenteeism and resource shortages than public schools in developing countries.

Research on the relative efficiency of private and public schools in developing countries reaches mixed conclusions. Some studies report that private schools are more cost effective than public schools in developing countries, specifically due to gross inefficiencies in the public system that are tied to corruption, bribery, inefficient spending, the existence of ghost public schools, nepotism in teacher appointments and transfers, higher salaries, and the aforementioned teacher absenteeism (Bold et al., 2013; Chaudhury et al., 2006; Tooley et al., 2011; Transparency International, 2013). In contrast, a recent comparative study of school efficiency in Ethiopia, India, Peru, and Vietnam finds that public schools are more efficient than private schools in producing scores on a vocabulary test for most student subgroups (Johnes & Virmani, 2020). When statistical controls for family wealth and educational expenditures are removed from the analysis, however, private schools demonstrate an efficiency advantage. Key lessons from this literature are the necessity to control effectively for selection factors when comparing the outcomes from private and public schools and the importance of assessing heterogeneity in impacts due to cross-country differences.

Two voucher programs had special program characteristics. The initiative in Colombia has used student performance incentives, so the results for Bogota, Colombia, represent a blend of student incentives, additional education spending through top up by parents, and private school effects. The Colombia program has also used different tests in Outcome Years 3 and 7. The Louisiana voucher program contained test-based accountability provisions, using criterion referenced tests, not found in any of the other programs in our study. Since the curricula used in Louisiana private schools may not be aligned to that test to the degree they are in Louisiana public schools, the treatment effects may be influenced by the content and format of the test. A dichotomous variable accounts for the special characteristics associated with the two programs in Louisiana and Colombia.

While most voucher programs are evaluated for only a few years, the relationship between years of treatment and impact may be non-linear. This concern creates a dilemma for drawing policy conclusions. Conclusions drawn from the early years of an evaluation, though easier and cheaper to produce, may be meaningfully different from conclusions drawn from later years, which are more policy relevant as they allow for adjustments by the school or family in response to the voucher (Das et al., 2013; Glewwe & Muralidharan, 2016). Thus, years treated is a crucial moderator to study.

We set clear inclusion and exclusion criteria, selected only rigorous experimental studies, and coded moderators to test for heterogeneity in impacts and subgroup analyses. Yet, many other factors could account for remaining heterogeneity in impacts. Lack of information across the studies has prevented us from coding the possible moderators of test rigor and reliability, use of instructional time, regulations, implementation fidelity, religious orientation, and other factors.[6]

Effect size choice and calculations

Meta-analysis combines results from studies that individually have relatively small sample sizes and low precision. The effect sizes have been standardized to allow comparability across studies. We estimated Hedges' g and its standard error (Hedges & Olkin, 1985; Appendix 1). To minimize human error and to ensure coding reliability, two team members separately extracted the information necessary to compute the effect sizes and standard errors. The two members discussed any remaining discrepancies with the third team member to fully agree on the data extraction and assumptions, resulting in an inter-rater reliability of 100%.

Some studies only provide subgroup impacts. Impacts for African Americans and Other Ethnic Groups are provided in Peterson et al. (2003), and impacts for local languages and English are provided in Wolf et al. (2015) and Muralidharan and Sundararaman (2015). In a fixed-effects meta-analysis, one true effect size is assumed to be shared by the studies, including effect sizes drawn from subgroups in a study. We have conducted a fixed-effects meta-analysis of subgroup impacts to compute overall impacts for inclusion in our main analysis. We have never combined impacts across treated years, policy parameters (TOT/ITT), or reading/math. The final meta-analysis contains 144 effect sizes that are actual study impacts (34 reading TOT estimates, 35 math TOT estimates, 37 reading ITT estimates, and 38 math ITT estimates).

Checking for robustness to program attrition

Sample attrition challenges our confidence in a study's ability to estimate the true program impact. Therefore, we have categorized the program impacts as either "low" or "high" risk of attrition bias based on Cochrane's risk of bias tool (Higgins & Green, 2011).[7] For multiple studies that analyzed the same sample, if one of the studies was labeled low risk, the other studies have been read to check if they contradicted this classification. If no discrepancy was found, the overall impacts for a particular program have been labeled low risk for attrition bias. Only the impacts for Delhi, India, were labeled high risk for attrition bias, as the authors did not carry out any checks for sensitivity of the results to sample attrition. We thus repeated the meta-analysis by excluding the impacts for Delhi.[8]

Selection of empirical models

Random effects

Due to the heterogeneous contexts of our studies, we use a random effects meta-analysis model to create an overall effect size by combining the effect sizes extracted from each study. The random effects model assumes that effect sizes are independent and randomly drawn from a single effect-size distribution (Borenstein et al., 2007). The study-specific effect sizes have been weighted by the inverse of their variance. The between-study heterogeneity has been incorporated in the random effects. The weighted average effect size $\overline{\mathrm{ES}}$ and its sampling variance $V(\overline{\mathrm{ES}})$ in the random effects model is described by the equations

$$\overline{\mathrm{ES}} = \frac{\sum_{i=1}^{m} (w_i * \mathrm{ES}_i)}{\sum_{i=1}^{m} w_i} \quad \text{and} \quad V(\overline{\mathrm{ES}}) = \frac{1}{\sum_{i=1}^{m} w_i}$$

where m is the number of studies, a study i's weight is $w_i = 1/(\nu_i + \tau^2)$, ν_i is the within-study variance (a known value), and τ^2 is the between-study heterogeneity from a method of moments estimator. The sampling error of the average effect size is obtained from its sampling variance.

Robust variance estimation (RVE)

Several inter-program dependencies in the data are not reported in the studies. The dependencies among effect sizes are hierarchical, as 144 effect sizes are clustered within 21 studies that represent 11 programs. Such dependencies are among math and reading impacts, impacts across different studies, or samples, and impacts across treated years due to individual students contributing to multiple effect sizes. RVE models dependencies when the within-study covariance is unknown (Hedges et al., 2010). We repeated the meta-analyses with RVE based on hierarchical effects using small-sample corrections, as the degrees of freedom are limited and covariates are unbalanced for several subgroups (Tipton, 2015). Though RVE is methodologically superior to random effects, as the former addresses dependencies in the data, the robustness of effect sizes across the two empirical models should increase our confidence in the results.

Meta-regression

We explore impact heterogeneity by incorporating known study features as moderators in a meta-regression model. For random effects meta-regression, the equation assumes that the study effect sizes ES_i for a study i are a function of j moderators:

$$\mathrm{ES}_i = \beta_0 + \beta_1 X_{i1} + \beta_1 X_{i2} + \ldots + \beta_j X_{ij} + \phi_i + \epsilon_i$$

$$\epsilon_i \sim N(0,\ \nu_i)$$

$$\phi_i \sim N(0,\ \tau_c^2)$$

where τ_c^2 is the residual unexplained variation (method-of-moments estimator) and ν_i is the residual sampling error (assumed to be known) for study i. In the above equation, ϕ_i and ϵ_i are independent and β are estimated using weighted least squares with inverse variance weights.

To account for data dependencies, we ran the meta-regression with RVE. The coefficients obtained are consistent estimates of the underlying population parameters (Fisher & Tipton, 2015). The results are consistent even under non-normality, and the model does not require the explanatory variables to be fixed, as with random effects meta-regression. The covariate for Years of Treatment varies both within and between the 11 program clusters. To distinguish the within and between program cluster effects of this covariate, we centered it (Fisher & Tipton, 2015; Tanner-Smith & Tipton, 2014). This transformation resulted in a program mean and a program centered mean for Years of Treatment.[9]

Results

We present the overall results for the last year of each program and separate results for US and international locales in Table 4. We use effect sizes across all years to analyze impacts for the full sample and by key program moderators in Table 5. Results from random effects and RVE models are presented side by side. The effect sizes are shown in standard

Table 4. Meta-analytic impacts based on last year of a program treatment.

	Random effects			Robust Variance Estimation				
Sample	Pooled ES	*SE*	*N*	Pooled ES	*SE*	N_s	N_E	*df*
Overall	0.126***	0.043	56	0.126**	0.064	11	56	7.3
Reading	0.160**	0.070	27	0.160*	0.083	10	27	6.8
Math	0.094*	0.050	29	0.094*	0.052	11	29	7.5
USA								
Reading	0.047**	0.020	21	0.050**	0.025	7	21	3.3
Math	0.043*	0.024	23	0.045	0.035	8	23	4.1
International								
Reading	0.459**	0.184	6	0.459	0.296	3	6	2
Math	0.217	0.150	6	0.217	0.197	3	6	2

Note: Results are based on random effects and robust variance estimation with hierarchical effects. Pooled ES = the weighted average of effect sizes in this category; *SE* = standard error; Last Year impacts are for a program. DC Opportunity Scholarship Program has two distinct program samples. Due to low degrees of freedom for program moderators, analysis by other program moderators is not carried out for last year of a program. N_s = number of programs; N_E = number of effect size estimates; N = number of effect sizes; df = degrees of freedom. Robust variance estimation is used to cluster standard errors within programs. Small sample corrections are used. When $df < 4$, Tipton (2015) notes that the normal approximation fails and p values should be interpreted with caution.
***$p < 0.01$. **$p < 0.05$. *$p < 0.1$.

Table 5. Meta-analytic impacts based on all effect sizes.

	Random effects			Robust Variance Estimation				
Sample	Pooled ES	*SE*	*N*	Pooled ES	*SE*	N_s	N_E	*df*
Full	0.064***	0.020	144	0.064*	0.035	11	144	5.6
Subject								
Reading	0.079***	0.030	71	0.079*	0.041	10	71	5.5
Math	0.048*	0.025	73	0.049	0.038	11	73	5.6
Location								
International	0.205***	0.068	24	0.205	0.147	3	24	2
USA	0.028**	0.012	120	0.029	0.025	8	120	3.5
Policy Parameters								
TOT	0.053	0.036	69	0.053	0.052	10	69	5
ITT	0.072***	0.018	75	0.072***	0.026	10	75	4.9
Funding Type								
Private	0.052***	0.011	88	0.052***	0.016	7	88	2.5
Public	0.072	0.051	56	0.072	0.091	4	56	2.4
USA Funding Type								
Private	0.046***	0.010	72	0.048**	0.021	5	72	1.5
Public	−0.005	0.025	48	−0.004	0.062	3	48	1.7

Note: Results are based on random effects and robust variance estimation with hierarchical effects. Pooled ES = the weighted average of effect sizes in this category; *SE* = standard error; DC Opportunity Scholarship Program has two distinct program samples. N_s = number of programs; N_E = number of effect size estimates; *N* = number of effect sizes; *df* = degrees of freedom. Robust variance estimation is used to cluster standard errors within programs. Small sample corrections are used. When $df < 4$, Tipton (2015) notes that the normal approximation fails and *p* values should be interpreted with caution.
***$p < 0.01$. **$p < 0.05$. *$p < 0.1$.

deviation (*SD*) units. Effect sizes are similar across the two empirical models, but the RVE method produces larger, more conservative, standard errors. When the degrees of freedom are below 4 in the RVE model, the results should be viewed as suggestive as the *p* values are not reliable (Tipton, 2015).

Overall impacts from the last year of a program

Voucher impacts may change in sign, magnitude, and statistical significance across years treated. Analyses based on the most recent year are considered more policy relevant than earlier results, as stakeholders may have changed their behavior in response to vouchers (Das et al., 2013; Glewwe & Muralidharan, 2016). The effects from the last study year represent the cumulative effects of the school voucher intervention across all evaluation years. Overall, vouchers produced an achievement impact of 0.126 *SD* that is statistically significant with 95% confidence intervals (CI) in the RVE model and 99% in the random effects model. The point value of the impact is higher in reading (0.160 *SD*) than in math (0.094 *SD*), and statistically significant with 90% CIs in both meta-analytic models. The separate US and international impacts are only suggestive ($df < 4$ in RVE), but show that reading impacts may be higher than math impacts, particularly for the non-US programs. The US math impacts (0.045 *SD*) are statistically null in the RVE model but significant with 90% CIs in the random effects model. Other program moderators' analysis is not carried out for the final program year due to low degrees of freedom.

To address concern that by focusing on the last year of each program available, we might be only including relatively successful programs, we conduct further checks described in the sections on impacts for full sample and program moderators (including effect sizes for all evaluation years) and meta-regression (in which a meta-regression confirms a timing effect). We also assess the possibility of publication bias (using a funnel plot and fail-safe *N*).

Impacts for full sample and program moderators

The data on program impacts are asymmetrical across the 11 programs. The programs have been evaluated for different numbers of years, and impacts in the final year of analysis are the only performance measure we have for all the programs. Analysis based only on effect sizes from the last year of a study may exaggerate their consistency. Results in Table 5 use effect sizes for all evaluation years. The full sample yields an impact of 0.064 *SD* that is statistically significant with 99% CIs in the random effects model but only with 90% CIs in the RVE model. Its magnitude is half the effect size based on the last year impacts. Voucher impacts display a timing effect, growing more positive over time. Across program years, vouchers produce a higher impact in reading (0.079 *SD*) than in math (0.048 *SD*). Suggestive results ($df < 4$ in RVE) by location show a larger achievement effect internationally (0.205 *SD*) than in the US (0.029 *SD*).

Our reliance on the final year impact estimates or multiple annual estimates from longitudinal studies might be positively biasing our meta-analysis. Pilot programs may be closed or evaluations ended prematurely if the initial study findings are negative or otherwise disappointing, thereby producing a distribution of observed impacts truncated based on initial program effectiveness.[10] To our knowledge, all 11 of the programs in our study continue to operate, but 8 of the 21 evaluations that inform the meta-analysis ended after 1 or 2 years (Table 2). If we restrict our analysis to findings after 1 year of program treatment, our estimates of the test score effects of school vouchers are all positive but not statistically significant. A typical student who switches from public to private school with a voucher, however, would experience a school-switching penalty in that initial year of approximately −0.05 *SD* in math achievement and a similar loss in reading achievement (Hanushek et al., 2004). The 2nd-year achievement effects from voucher experiments arguably are a more reliable measure of initial program effects. Those effects average essentially zero in math across the studies in our meta-analysis but average a gain of 0.05 *SD* in reading that is statistically significant.[11] Thus, our findings of statistically significant positive achievement effects of vouchers in reading in our main analysis do not appear to be due to a truncation in the quality distribution of the programs being studied, although that phenomena could affect our math findings.

The impacts by type of policy parameter show that the mere offer of a voucher (ITT) has an achievement impact of 0.072 *SD* that is statistically significant with 99% CIs in both meta-analytic models. The impact of using a voucher (TOT) is statistically null, partly because the subsample of studies[12] with TOT estimates has produced less positive findings than the ITT subsample and partly because some TOT adjustments introduce noise into the analysis. Privately funded programs yield an impact of 0.052 *SD* that is significant with 99% CIs, whereas publicly funded programs yield a statistically null impact. The results are merely suggestive as $df < 4$ in RVE. A similar pattern is seen for privately and publicly funded programs in the US. Due to low degrees of freedom, we refrain from presenting results for even smaller subgroupings of studies.

Meta-regression

Meta-regression incorporates the effect sizes along with study moderators in a regression framework. The study moderators are the independent variables, whereas the effect sizes are the dependent variable. We present meta-regression results based on random effects and RVE models side by side. The degrees of freedom are very low, so the results are merely suggestive.

Results in Table 6 are consistent with the timing effect evident in Tables 4 and 5. The longer a sample of voucher students has been treated, the larger and more positive the achievement effects tend to be. Privately funded programs again show a higher impact than publicly funded ones. Other results lack statistical significance but hint at larger impacts in reading than math and outside the US than within, consistent with Tables 4 and 5.

Heterogeneity analyses

The proportion of between-study variance explained for the random effects model is 37.7%, whereas the percent of residual variation due to heterogeneity is 90.2%. The effects may appear more consistent than they are because not all study differences are observable. In Tables 4 and 5, we have explained some of the variation in program impacts through subgroup analysis. Given our limited sample of 11 voucher programs,

Table 6. Meta-regression.

Variables	Random Effects ES	Robust Variance Estimation ES	Robust Variance Estimation *df*
Reading	0.034	0.034	5.5
	(0.027)	(0.038)	
TOT	0.001	−0.000	4.4
	(0.029)	(0.020)	
PM_YearsofTreatment	0.139***	0.135	2.7
	(0.025)	(0.112)	
PCM_YearsofTreatment	0.063***	0.062*	4.6
	(0.014)	(0.036)	
Privately funded	0.063*	0.063	2.7
	(0.036)	(0.059)	
International	0.018	0.021	1.7
	(0.047)	(0.120)	
Special Program Characteristics	−0.072	−0.072	2
	(0.056)	(0.278)	
Constant	−0.290***	−0.278	3.8
	(0.067)	(0.273)	
N_s		11	
N_E	144	144	

Note: Results are based on random effects and robust variance estimation with hierarchical effects. ES = the weighted average of effect sizes in this category; standard errors are in parentheses; DC Opportunity Scholarship Program has two distinct program samples. The covariate for Years of Treatment varies both within and between the 11 program clusters. To distinguish between the within program cluster and between program cluster effects of this covariate, we centered the variable (Fisher & Tipton, 2015; Tanner-Smith & Tipton, 2014). PM_YearsofTreatment provides estimate of between-program effects, whereas PCM_YearsofTreatment provides estimate of within-program effects for Years of Treatment.

N_s = number of programs; N_E = number of effect size estimates; *df* = degrees of freedom. Robust variance estimation is used to cluster standard errors within programs. Small sample corrections are used. When df < 4, Tipton (2015) notes that the normal approximation fails and *p* values should be interpreted with caution.

***$p < 0.01$. **$p < 0.05$. *$p < 0.1$.

the subgroup analysis leads to only suggestive results. Estimation of final-year effects show that the combined achievement effect, reading effect, and math effect all have differed significantly from each other. Estimates based on all years found that those same three effects (overall achievement, reading, and math) have differed significantly from each other, as did international from US, TOT from ITT, and privately funded from publicly funded. Additionally, the estimates vary slightly across studies of the programs that have been examined by multiple research teams. Such differences exist due to researchers' decision related to which baseline control variables are included in the estimation model, how missing data challenges are addressed, and even how students are classified by race (Krueger & Zhu, 2004; Peterson & Howell, 2004; Wolf & Egalite, 2017, p. 291).

Cost effectiveness

Studies find that vouchers are cost effective, since they tend to generate achievement outcomes as good as or better than traditional public schools at a lower cost (Muralidharan & Sundararaman, 2015; Wolf & McShane, 2013). Null voucher impacts obtained at a lower cost than public schooling may benefit society. The average per-pupil spending in the public school systems in our meta-analysis was higher than the voucher amounts.[13] Where the voucher covered the full cost of private school tuition, average school funding in the private versus the public sectors in the same geographic location where the voucher program was implemented was $255 versus $636 for Andhra Pradesh, India; $483 versus $1,963 for Delhi, India; $4,817 versus $11,846 for Milwaukee, WI; $5,456 versus $10,853 for Louisiana; $7,761 versus $21,081 for Washington, DC OSP I; and $12,306 versus $20,577 for Washington, DC OSP II.[14] The partial programs in Charlotte, NC; Dayton and Toledo, OH; NYC; Washington, DC; and Bogota, Colombia cost less than half of the per-pupil expenditures in the neighborhood public schools; however, private funds supplemented them. Across the US, in constant 2018–2019 dollars, the per-pupil spending in public schools averaged $14,241 in 2011–2012 while average private school tuition rates were $7,670 in Catholic schools and $9,670 in non-Catholic religious schools (National Center for Education Statistics, 2019a, 2019b).[15]

What do these differences in average student funding across the sectors mean in terms of the operation of public and private schools? The kinds of private schools that participate in school voucher programs tend to have more modest facilities and fewer special programs than local public schools. In their studies of the Dayton, OH, Washington, DC, and New York privately funded programs that inform this meta-analysis, Howell and Peterson (2006, pp. 94–95) report that the parents of students attending private schools through a voucher in all three cities were less likely than the parents of students in the control group to report that their child's school had a nurse's office, a separate cafeteria, a library, counselors, special programs for non-English speakers, and separate programs for students with disabilities. The parents of voucher students generally were more likely than parents of control-group students to say that their child's school offered a music program, an after-school program, and individual tutors. Parent surveys from studies of the full-tuition voucher programs in Washington, DC, and Milwaukee largely confirm these results (Witte, 2000; Wolf et al., 2007). Private school teachers in the US are paid 21% less than public school teachers, on average (Allegretto &

Tojerow, 2014), in part because they are less likely to have graduate degrees (Kisida et al., 2008). Unlike public schools, private schools are not required to provide free transportation to and from school, although some do. Generally, the kinds of private schools that serve low-income students in voucher programs are lean operations without the typical facility enhancements and customized programs provided at significant expense in the better funded public-school sector. In the few areas where a program or service is more common in private voucher schools than in public schools, such as after-school programs or tutoring, the enhancements tend to have a clear achievement focus. These contrasts between private schools serving voucher students and public schools in the US tend to be even greater in the developing world (Muralidharan & Sundararaman, 2015; Tooley, 2009; Wolf et al., 2015). Additionally, private school tuition rates (and related voucher amounts) may not cover the full per-pupil spending amounts, as other donations can contribute significant funds to school budgets, particularly in Catholic schools in the US (Gjelten, 2020).

Publication bias

Although we tried to reduce the possibility of publication bias, in order to strengthen our analysis we created a funnel plot for the full sample (Figure 2). Fewer studies with large *SE*s are present on the left side of the average pooled effect size in comparison to the right side, suggesting limited evidence of publication bias. An Egger's test (Egger et al., 1997) yields a bias coefficient of −1.387 that is statistically null. Publication bias is likely

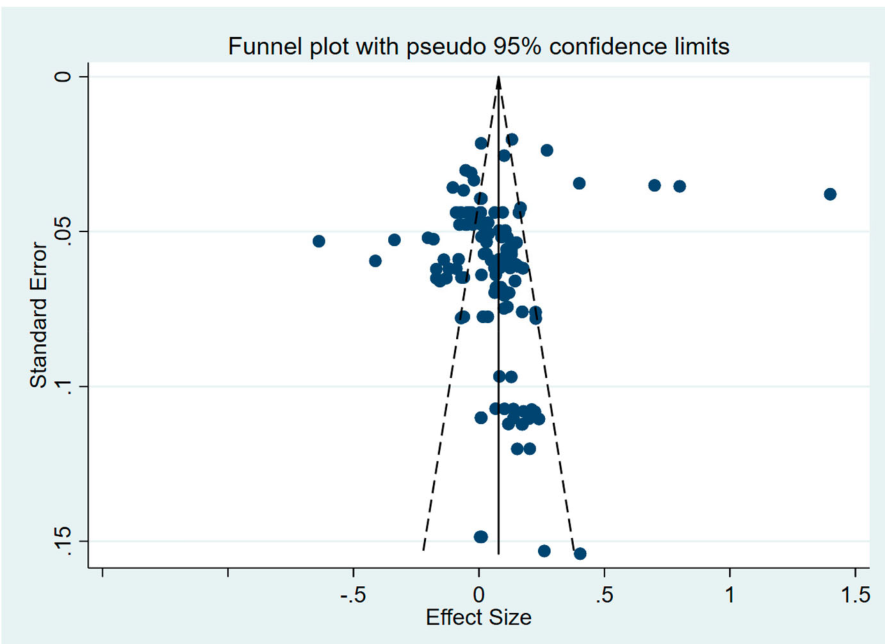

Figure 2. Funnel plot to assess publication bias in the meta-analysis.

Note: This funnel plot represents the Egger's test of the full sample (Egger et al., 1997). Egger's test, bias coefficient: −1.387, p = 0.172.

not present. We calculated a "fail-safe *N*" as an additional robustness check for our findings (Orwin, 1983; Rosenthal, 1979). The analysis showed that we would need to find 144 studies with no effect in order to drive down the meta-analytic effect size to half ~ 0.03 *SD*. Furthermore, in order to reduce the combined significance level of the meta-analytic estimate to a target alpha level of 0.05, we would need to find 10,882 null studies. Publication bias does not appear to be a concern for this meta-analysis.

Discussion

This meta-analysis contributes to our understanding of private school choice by combining rigorous evidence from all RCTs of the effects of voucher and voucher-type programs on student achievement. This review provides an up-to-date, methodologically coherent and comprehensive overview of all the rigorous experimental findings, and it yields important policy implications about the effectiveness of voucher programs. Our search process turned up 9,443 potential studies, 21 of which were included. These studies represent 11 programs, eight in the US and three in other countries. In total, 144 effect sizes are included, with 56 findings drawn from the last year of the studies. The results indicate generally modest positive achievement effects of school vouchers (ES = 0.126 *SD*) based on the last-year estimates from the various evaluations.

These results are heterogeneous. Longer term effects are larger than earlier estimates for the same program. Longer term effects are considered more policy relevant than immediate achievement effects, whenever available (Das et al., 2013; Glewwe & Muralidharan, 2016), and analyses comparing the size of the effects for the first 2 years, available from the authors by request, indicate that closure of programs or evaluation termination after a first year of negative or null findings do not appear to be driving these timing effects. Impacts are larger for programs outside the US. Privately funded programs yield a significantly positive impact, whereas publicly funded programs have a null impact.[16] The impacts are larger for reading than for math, although the difference is not statistically significant.

A meta-analysis of *all* education RCTs from 1995 to 2010 shows that the average impact of any education intervention on test scores is 0.08 *SD* at the elementary and 0.15 *SD* at the middle school level (Lipsey et al., 2012). Our meta-analysis finds that voucher interventions produce positive test score outcomes which are comparable in size to these benchmarks. The results indicate that voucher programs tend to moderately increase test scores, particularly in developing countries with a large private–public school quality gap. The results align with the Heyneman and Loxley (1983) hypothesis for developing countries that the quality of schools and teachers predominantly influence student learning.

Voucher effects on the students who remain in traditional public schools (TPS) also are important, but beyond the scope of this study. These systemic achievement effects of vouchers could be positive, due to institutional competition, or negative, due to resource losses in TPS. Voucher systemic effects have been positive in Louisiana (Egalite, 2014) and Milwaukee (Carnoy et al., 2007; Chakrabarti, 2008; Greene & Forster, 2002; Hoxby, 2003) but null in Washington, DC (Greene & Winters, 2007) and Andhra Pradesh (Muralidharan & Sundararaman, 2015). Jabbar et al. (2019) found positive competitive effects of US voucher programs in a meta-analysis. Modest average achievement effects of private school choice for participants are not coming at the cost of test score declines for non-choosing students (Egalite & Wolf, 2016; Forster, 2016).

Limitations

Although the scope of our search process was global, more international RCTs are needed to reach definitive conclusions about the impacts of voucher programs around the globe. Our search process yielded RCTs on private school vouchers only in the US, India, and Colombia. With only 11 voucher programs and limited variation in program moderators, we cannot fully understand the underlying dynamics that might affect program outcomes.

Implications for future research

Many of the programs included here are still relatively small scale. More experimental work should take place on larger programs in order to understand whether or not the achievement benefits of private school vouchers replicate at scale. We cannot learn much from even large-scale programs if they are not implemented alongside an experimental evaluation (Muralidharan & Niehaus, 2017). Further, more experimental evaluations that consider the impacts of vouchers on non-test score outcomes such as educational attainment and civic values (e.g., Angrist et al., 2006; Wolf, 2007; Wolf et al., 2013) would be of great value to the field. We hope that our study motivates researchers to pursue experimental evaluations of voucher impacts whenever feasible.

Notes

1. After we had submitted our study for review, five studies came out as conference presentations, working papers, and publications in 2019–2020. These studies analyzed the 3rd year of the second evaluation of the DC Opportunity Scholarship Program (Webber et al., 2019), the 4th year of the Louisiana Scholarship program (Mills & Wolf, 2019), the 3rd- and 4th-year analysis of the ENABLE voucher program in Delhi, India (Dixon et al., 2019), a reanalysis of the same ENABLE voucher program (Crawfurd et al., 2019), and a voucher program in Uganda (Barrera-Osorio et al., 2020). These five studies were reported too late to be included in our meta-analysis. Since four of the five new experimental evaluations are extensions or replications of studies already in our sample, and three of them report generally more positive/less negative results while the other two report less positive/more negative results, we doubt that adding these recent findings to our analysis would change our results meaningfully.
2. The newest form of private school choice in the US, education savings accounts (ESAs), permits parents to secure educational services from multiple private providers (Butcher & Burke, 2016). Other programs, in the US and globally, fund scholarships through private donations with no special tax incentive. This study does not cover ESAs, because no evaluations of their effects on student outcomes exist. We do include privately funded scholarships, tax-credit scholarships, and school vouchers because they accomplish the same general purpose.
3. Information for effect sizes in English and local languages were coded for international studies. The reading estimate for Delhi, India, includes an overall estimate for English and Hindi. The reading estimate for Andhra Pradesh, India, includes an overall estimate for English, Hindi, and Telugu. The reading estimate for Bogota, Colombia, is for Spanish.

Parents in developing countries assign high importance to English during the school selection process due to its likely association with increased academic opportunities and a higher economic return from the job market (Azam et al., 2013; Mitra et al., 2003; Sen & Blatchford, 2001; Tooley & Dixon, 2003).

4. Muralidharan and Sundararaman (2015) had a two-stage randomization and it also randomized at village level.
5. The Washington Scholarship Fund (WSF) was a privately funded scholarship program that preceded the government-funded Opportunity Scholarship (voucher) Program (OSP). The WSF was phased out in 2009, five years after the OSP was launched.
6. We also did no code setting (rural or urban) and region within a country as these variables do not further distinguish among the 11 voucher programs, above and beyond the moderators already included.
7. We classified individual studies as "low" risk for bias if they either met the What Works Clearinghouse (WWC) standards for acceptable overall and non-differential sample attrition or passed an appropriate robustness test for sample attrition bias. Robustness tests included testing for baseline equivalence between treatment and control group students post-attrition, analyzing the sensitivity of the estimates to artificial truncations in the respondent samples, and various bounding analyses such as inverse probability weighting or Lee bounds (D. S. Lee, 2009) to check for the robustness of the results to differential attrition.
8. Removing the estimates for Delhi does not affect the overall final year results. It merely increases the subgroup reading impact for international studies from 0.459 *SD* to 0.625 *SD*.
9. The meta-analysis was carried out in STATA 15 using the metan command with random effects, robumeta package with hierarchical effects, metareg command for a random effects meta-regression, and metafunnel and metabias for assessing publication bias. We conducted additional publication bias checks in R using R studio environment.
10. This concern is related to the "publication bias problem" in many meta-analyses, whereby studies with null findings tend to end up filed away instead of being released into the public domain, biasing the set of available studies towards those with statistically significant positive or negative findings.
11. Detailed results discussed in this paragraph are available from the lead author by request.
12. As stated before, ITT impacts were not available for all studies (e.g., the Louisiana Scholarship Program was a placement lottery, and it has only TOT effects).
13. The ideal cost comparison would be average total per-pupil expenditure in local district-run public schools compared to average total expenditure per voucher student in a given voucher program. Except in rare cases, neither figure is reported publicly nor can be calculated based on public information. Public school districts exclude large categories of expenditure – such as capital costs, contributions to teacher pensions, and transportation costs – from the per-pupil totals before they report them publicly as "foundation funding" (DeAngelis, Wolf, et al., 2020). Parents and private school internal financial aid programs often supplement voucher amounts. Thus, both the average per-pupil expenditure in local public schools and the maximum voucher amount in voucher programs are underreports of the actual amount of resources expended on students. A rough comparison of one undercount to another undercount is the best that can be provided under these circumstances.
14. All amounts are inflation-adjusted 2013 dollars based on the last year of program evaluation. Weighted averages are taken across cohorts in Washington, DC OSP II. For India, rupees have been converted to US dollars.
15. Most school voucher enrollments in the US are in Catholic or non-Catholic religious schools. The most recent year in which private school average tuition rates were available from the National Center for Education Statistics was 2011–2012.
16. Privately funded private school choice programs tend to differ from publicly funded ones in several important respects. They tend to have higher ceilings on income eligibility and permit participating private schools to require family payments to supplement the cost of the voucher, which typically only covers part of the tuition. Thus, privately funded programs serve a population of students who are less disadvantaged economically and socially than

students in publicly funded programs. Privately funded programs impose fewer regulations on participating private schools than do publicly funded programs. Survey experiments suggest that some voucher regulations discourage private schools from participating in voucher programs, especially if the school is higher quality based on various metrics (DeAngelis, Burke, et al., 2020; DeAngelis et al., 2019). These differences underscore a tension among policy makers between ensuring that a private school choice program delivers an effective programmatic treatment and ensuring that the treatment is targeted to needy students.

Acknowledgements

Earlier versions of this paper were presented at conferences of the Society for Research on Educational Effectiveness, Association for Education Finance & Policy, and Association for Public Policy Analysis & Management International Conference. We thank Anna Egalite of North Carolina State University and Jonathan Mills of University of Arkansas for details regarding studies included in our analysis. We also thank Mark Lipsey of Vanderbilt University, Philip Gleason of Mathematica Policy Research, John Witte of University of Wisconsin, and Gary Ritter of Saint Louis University for comments on earlier drafts. Finally, we thank Elizabeth Tipton of Northwestern University and Katharine Conn of Teachers College Columbia University for guidance on statistical methods used in this study. We also thank Kathleen Wolf of University of Arkansas for editorial review of the manuscript. We own all remaining flaws.

Disclosure statement

No potential conflict of interest was reported by the authors.

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Appendix 1. Formulas used for effect size choice and calculations

1. Mean differences: $\bar{X}_T - \bar{X}_C$
2. *SD* Pooled: $\mathrm{Std}_{(pool)} = \sqrt{\dfrac{S_1^2(n_1 - 1) + S_2^2(n_2 - 1)}{n_1 + n_2 - 2}}$
3. Cohen's *d*: $d = \dfrac{\bar{X}_T - \bar{X}_C}{Std_{(pool)}}$
4. Lower bound ES (95%): LB = ES − ($SE_d * 1.96$)
5. Upper bound ES (95%): UB = ES + ($SE_d * 1.96$)
6. Effect Size by correlation: $ES = \dfrac{2r}{\sqrt{1 - r^2}}$
7. Effect Size by *t* ratio: $d = t\sqrt{\dfrac{n_1 + n_2}{n_1 n_2}}$
8. Hedges' *g* (Unbiased *d*): $ES(d') = \left[1 - \dfrac{3}{4N - 9}\right]d$
9. Standard error for effect size: $SE_{d'} = \sqrt{\dfrac{n_1 + n_2}{n_1 n_2} + \dfrac{d'^2}{2(n_1 + n_2)}}$

10. Inverse Variance (w) $w = \frac{1}{(SE)^2}$

11. Grand Effect size: $\overline{\mathrm{ES}} = \frac{\sum (w \times \mathrm{ES})}{\sum w}$

where ES is effect size of each study, *SE* is its associated standard error, and *w* is the inverse variance weight.